

Shared Substring Alignment

Identifying recurring structure within the encrypted GPS special messages

Group B — 10 shared characters, same day

2014-10-08	T	3	"	E	H	S	D	S	"	I	4	H	.	'	H	X	-	+	7	U	X		PRN 32
2014-10-08	"	J	Y	3	2	Q	8	/	S	'	°	5	.	'	H	X	-	+	7	U	X		PRN 29

Group D — 9 shared characters, 9 months apart

2021-06-05	Q	"	O	:	T	K	L	Y	4	7	I	R	P	1	6	A	C	0	2	4	B	A	PRN 31
2020-09-19	O	H	6	°	S	3	L	Y	4	7	I	R	P	1	6	K		X	D	.	4	L	PRN 30

Group A — 8 shared characters, 3 weeks apart

2019-12-11	2	C	K	0	E	6	:	U	'	5	P	S	F	8	T	J	9	0	F	D	X	'	PRN 1
2019-11-20	Q	°	5	H	Y	1	:	U	'	5	P	S	F	8	T	7	8	J	9	7	I	Y	PRN 31

Group C — 7 + 4 shared characters, 4 days apart

2024-04-07	4	3	P	A		:	Z	U	°	S	G	Z	8	P	J	D	L	-	:	I	M		PRN 3
2024-04-11	:		2	N	K	M	Z	U	°	S	G	Z	8	P	J	E	G	B	:	I	M		PRN 3

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22

Byte position

Identical character at same position

Differing character